

Targetless Spatiotemporal Calibration of 3D Radar-Camera Pairs in Continuous Time: a Python re-implementation and reproduction of Wise, Cheng & Kelly (IEEE T-RO 2023)

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re-implementation, simulation studies and handheld-rig front end, September 2026

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Abstract

A 3D millimetre-wave radar measures its own ego-velocity from the Doppler range-rates of static scatterers; a monocular camera measures its pose up to scale with SLAM. Wise, Cheng and Kelly [1] fuse the two streams on a continuous-time B-spline trajectory to estimate the radar-camera extrinsics $\mathbf{T}_{cr}$, the camera scale α and the temporal offset τ without targets or overlapping fields of view. This document records a from-scratch Python re-implementation (cumulative cubic B-splines on $\mathbb{R}^3 \times SO(3)$ with analytic body velocities, checkerboard/PnP camera measurements, a closed-form initialiser derived from the paper’s identifiability model, and a sparse Levenberg–Marquardt joint estimator with a finite-difference Jacobian), the reproduction of the paper’s simulation studies on two trajectories and four noise levels (100 trials each), a numerical check of the identifiability analysis, and the radar ego-velocity front end run on the paper’s own 2020 handheld-rig recording (Zagreb corridor), which gives an inlier range-rate spread of 0.031 m/s, the “lower end” of the 0.05–0.15 m/s range the paper assumes. On the low-linear/high-angular-velocity trajectory the median errors at the lowest noise are 0.3 cm, 0.55°, 0.43 % and 1.7 ms; on the high-linear/low-angular trajectory translation and offset are, as in the paper, several times less well determined, and at three times the radar noise the scale degrades to 3–5 % because the simulated rig moves at well under 1 m/s. The full spatiotemporal solve on the real data was not re-run because it needs ORB-SLAM3 camera poses, which are not available here.

Keywords: radar-camera calibration, temporal calibration, continuous-time estimation, B-splines, ego-velocity, identifiability.

1 Overview

The paper’s code (Ceres, ORB-SLAM3, the REVE ego-velocity estimator) is not public. Everything here is a clean Python re-implementation in `experiments/`: `rc.py` (spline, simulator, initialiser, estimator, observability matrix), `sim_studies.py` (paper §V, sharded), `real_frontend.py` (paper §VI-A on the handheld bag), `make_tables.py` and `make_video.py`.

2 Method

2.1 Continuous-time trajectory

The camera pose $\mathbf{T}_{wc}(t) = (\mathbf{R}_{wc}(t), \mathbf{p}_c(t))$ is a uniform cumulative cubic B-spline (Sommer et al. [3]; paper eqs. 2–6) with knot spacing $\Delta t = 0.1$ s: with $u = (t - t_i)/\Delta t$ and cumulative basis $\lambda_j(u)$, $\mathbf{p}(u) = \mathbf{p}_i + \sum_{j=1}^3 \lambda_j(u)(\mathbf{p}_{i+j} - \mathbf{p}_{i+j-1})$ and $\mathbf{R}(u) = \mathbf{R}_i \prod_{j=1}^3 \exp(\lambda_j(u) \log(\mathbf{R}_{i+j-1}^\top \mathbf{R}_{i+j}))$. The body angular velocity is evaluated analytically, $\boldsymbol{\omega} = \dot{\lambda}_3 \mathbf{w}_3 + \mathbf{A}_3^\top (\dot{\lambda}_2 \mathbf{w}_2 + \mathbf{A}_2^\top \dot{\lambda}_1 \mathbf{w}_1)$, and both $\dot{\mathbf{p}}$ and $\boldsymbol{\omega}$ were verified against central differences to 10^{-9} .

2.2 Measurements

Camera at t_k : $\tilde{\mathbf{R}}_k = \mathbf{R}_{wc}(t_k) \exp(\mathbf{n}_r)$, $\tilde{\mathbf{p}}_k = \alpha \mathbf{p}_c(t_k) + \mathbf{n}_t$ (paper eqs. 12–13). Radar at radar time t_j (camera time $t_j + \tau$):

$$\mathbf{h}_j = \mathbf{R}_{cr}^\top (\mathbf{R}_{wc}^\top \dot{\mathbf{p}}_c + \boldsymbol{\omega}_c \times \mathbf{r})|_{t_j+\tau} + \mathbf{n}_v, \quad (1)$$

where $\mathbf{r}$ is the radar position in the camera frame; this is the paper’s eq. 10 written with the camera as the spline body. Residuals: $\mathbf{e}_r = \log(\tilde{\mathbf{R}}_k^\top \mathbf{R}_{wc}(t_k))$, $\mathbf{e}_t = \tilde{\mathbf{p}}_k - \alpha \mathbf{p}_c(t_k)$, $\mathbf{e}_v = \mathbf{h}_j - \hat{\mathbf{h}}_j$, weighted by $1/0.1^\circ$, $1/5$ mm and $1/\sigma_r$ (the PnP noise measured in simulation is 0.11° and 2.4 mm).

2.3 Initialisation

The paper’s identifiability model (eq. 21), $\mathbf{a}(t) = \alpha \mathbf{R}_{cr} \mathbf{h}(t + \tau) - \alpha \boldsymbol{\omega}_c \times \mathbf{r}$ with $\mathbf{a} = \mathbf{R}_{wc}^\top \dot{\mathbf{p}}$ the scaled camera velocity from a spline fitted to the camera poses alone, is linear in $\text{vec}(\alpha \mathbf{R}_{cr})$ and $\alpha \mathbf{r}$ for a fixed τ : the re-implementation solves this 12-parameter least-squares problem on a grid of $\tau \in [-0.2, 0.2]$ s, picks the residual minimum, takes α as the mean singular value of the 3×3 block and $\mathbf{R}_{cr}$ as its $SO(3)$ projection. Over the simulation studies this closed form starts within 1.3° (A) and 1.3° (B) median rotation error (Table 2); no ground-truth-based initialisation is used.

2.4 Joint estimator

Unknowns: n translation control points, n rotation control points (as local perturbations $\mathbf{R}_i^0 \exp(\boldsymbol{\delta}_i)$, re-linearised after every accepted step), $\mathbf{R}_{cr}$ (rotation vector), $\mathbf{r}$, α , τ : $6n + 8$ parameters ($n \approx 300$ for 30 s). Levenberg–Marquardt with a finite-difference Jacobian evaluated through the sparsity pattern (57 colour groups; step 10^{-6} , 10^{-4} s for τ), sparse LU on the normal equations, damping λ adapted by $3 \times / 5 \times$. The covariance of the eight global parameters is read from the final Gauss–Newton information. One solve takes about 9 s on a loaded laptop core.

2.5 Identifiability

The paper’s observability matrix $\mathbf{O}$ (eqs. 23–24) stacks $\partial \mathbf{h} / \partial [\mathbf{r}, \boldsymbol{\phi}, \alpha, \tau] = [-\alpha \boldsymbol{\omega}^\wedge, -\alpha (\mathbf{R} \mathbf{v})^\wedge, \mathbf{R} \mathbf{v} - \boldsymbol{\omega} \times \mathbf{r}, \alpha \mathbf{R} \dot{\mathbf{v}}]$ over time. Table 3 gives its singular values for trajectory B with full excitation, with rotation restricted to one axis, and at rest: one-axis rotation leaves rank 7 of 8, at rest the matrix vanishes, reproducing the paper’s §IV-C.

3 Experiments

3.1 Simulation studies (paper §V, Figs. 3–6)

Two analytic trajectories in front of a 9×7 checkerboard (0.1 m squares) at $z = 0$ (Fig. 1): A (high linear / low angular velocity, median 0.73 m/s, 0.39 rad/s) and B (low linear / high angular, 0.40 m/s, 1.75 rad/s), 30 s, camera 30 Hz, radar 20 Hz. Camera poses come from `cv2.solvePnP` on the projected corners ($f = 700$ px, 1280×1024) with pixel noise σ_c and a square size $\alpha = 0.5$ times the true one, so the measured positions carry the unknown scale; radar velocities from the analytic kinematics with noise σ_r ; $\tau = 50$ ms; extrinsics random ($\mathbf{R}_{cr} \sim \exp(\mathcal{N}(0, 0.8^2))$, $\mathbf{r} \sim \mathcal{N}(0, 8 \text{ cm})$) per trial. $(\sigma_r, \sigma_c) \in \{0.05, 0.15\} \times \{0.2, 0.4\}$, 100 trials each.

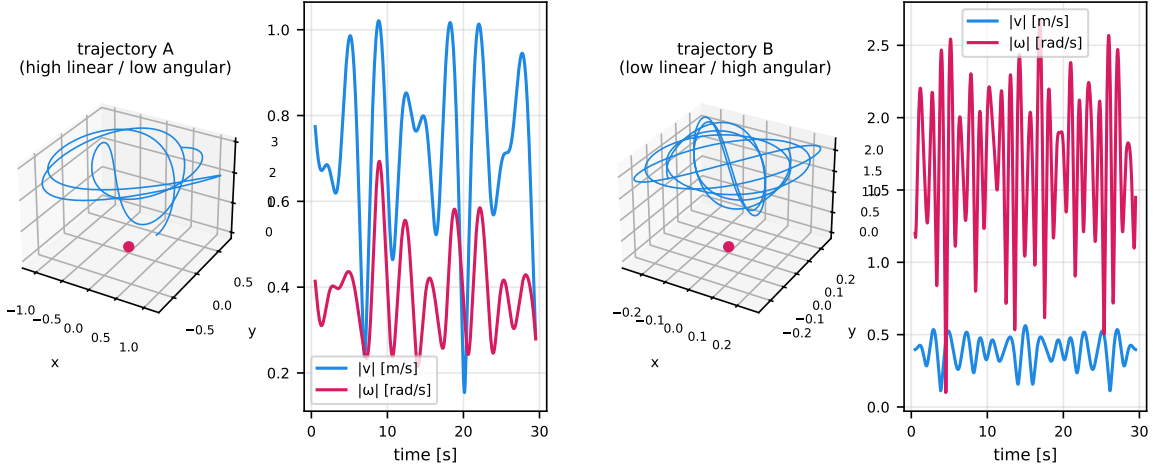


Figure 1: The two simulated trajectories and their speed / angular-rate profiles (paper Figs. 3–4).

Table 1: Calibration errors over 100 trials (median / 95th percentile / max), convergence rate and solve time.

traj.	σ_r	σ_c	translation [cm]			rotation [deg]			scale [%]			offset [ms]			conv.	time [s]
	[m/s]	[px]	med	p95	max	med	p95	max	med	p95	max	med	p95	max		
A	0.05	0.2	1.71	3.10	3.9	0.39	0.69	1.0	0.50	1.22	1.7	4.26	11.69	15.7	87%	21.6
A	0.05	0.4	2.58	4.53	5.6	0.39	0.81	1.0	0.90	1.80	2.2	3.81	11.26	14.3	91%	20.1
A	0.15	0.2	11.11	19.15	24.8	1.45	2.60	3.4	4.68	8.65	10.1	23.53	63.64	76.3	21%	32.6
A	0.15	0.4	13.64	21.25	25.0	1.42	2.95	4.1	4.89	7.42	8.3	25.09	66.14	112.8	49%	16.7
B	0.05	0.2	0.26	0.45	0.7	0.55	1.10	1.5	0.43	1.37	1.8	1.74	5.93	8.4	100%	8.2
B	0.05	0.4	0.27	0.52	0.6	0.49	1.00	1.3	0.43	1.17	1.8	2.08	5.23	10.7	100%	6.9
B	0.15	0.2	0.70	1.36	1.9	1.71	3.05	4.4	3.63	6.29	7.8	7.65	20.21	24.8	7%	12.8
B	0.15	0.4	0.75	1.36	2.0	1.46	2.98	3.4	2.84	5.26	7.1	9.83	24.60	58.3	41%	10.1

Table 1 and Figs. 2–3 reproduce the paper’s two qualitative results. Rotation is well determined on both trajectories at every noise level (medians 0.39–1.42° on A and 0.55–1.46° on B, the paper’s “less than two degrees”). Translation and temporal offset depend on the excitation: on B they stay at 0.3–0.7 cm and 1.7–9.8 ms (paper: within 10 cm and 10 ms), on A they widen to 1.7–13.6 cm and 4.3–25.1 ms with 95th percentiles of 21.2 cm and 66.1 ms at the highest noise (paper: “as large as 15 cm and 30 ms”). The scale is recovered to 0.50–0.43 % at $\sigma_r = 0.05$ m/s, the paper’s “one percent”, but to 2.84–4.89 % at $\sigma_r = 0.15$ m/s: the simulated rig moves at only 0.4–0.7 m/s, so a

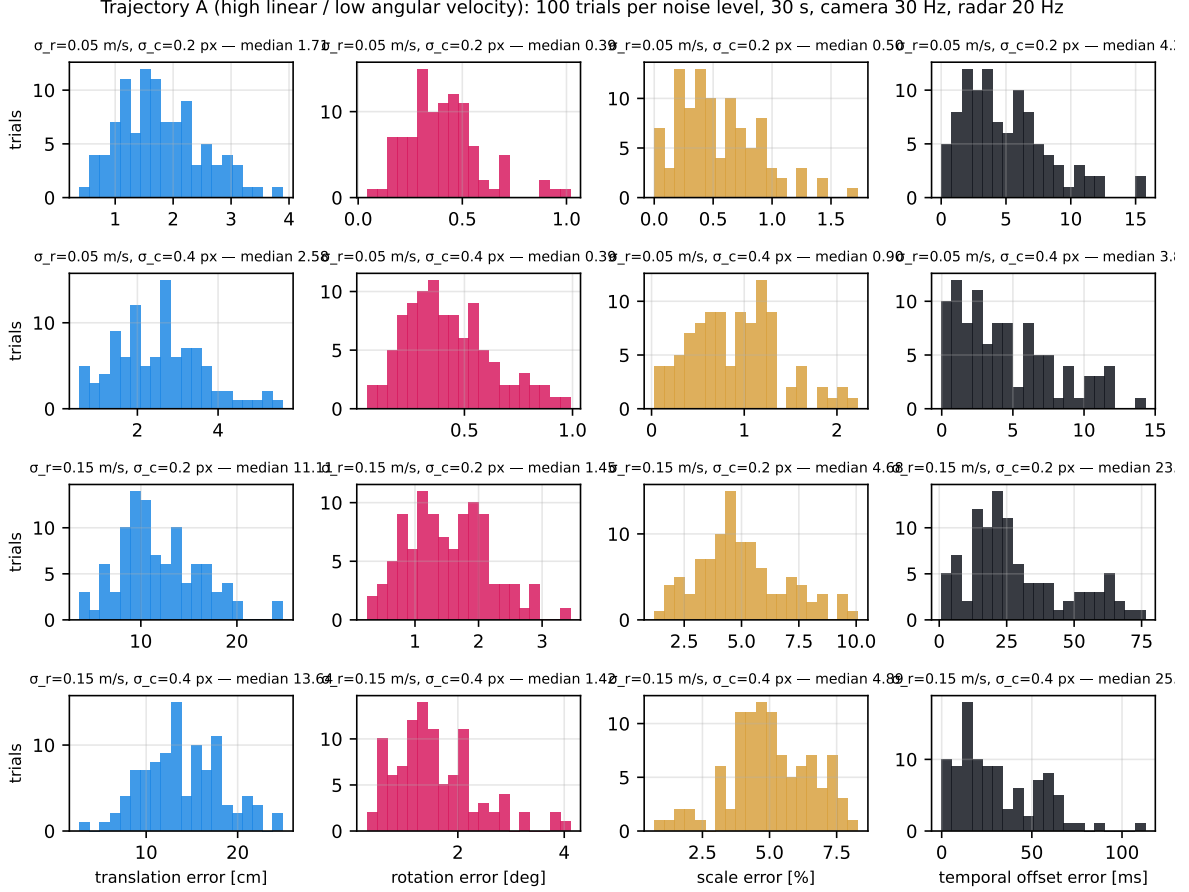


Figure 2: Trajectory A error histograms (paper Fig. 5).

0.15 m/s radar noise is a 20–40 % relative error on the very signal that carries the scale, whereas the paper’s trajectories (its Figs. 3–4) are faster. At $\sigma_r = 0.05$ m/s the estimator met its tolerance within 40 iterations in 87–100 % of the trials; at 0.15 m/s the cost surface is flatter and 7–49 % did (Table 1); the non-converged runs are slow tail iterations, not divergence, and are included in every statistic. Table 2 lists the initialiser’s error (median $1.3^\circ / 1.3^\circ$ over all settings) and the reported 1σ values, which track the empirical spread on B and under-report it on A (the flat directions of the information matrix are the ones the noise exploits).

3.2 Real data: radar ego-velocity front end (paper §VI-A)

The 2020 handheld recordings (TI AWR1843BOOST radar, Point Grey Flea3, VectorNav VN-100; corridors of the University of Zagreb FER building) were archived as ROS bags; the first 48-s segment is used. Each scan holds 23 returns (median) with (x, y, z) , intensity and Doppler; the 3-point RANSAC least-squares ego-velocity (paper eqs. 7–9, threshold 0.1 m/s, 300 hypotheses) succeeds on 80 % of the 594 scans with 21 inliers (83 %) per scan (Fig. 5). The inlier range-rate residual has a standard deviation of 0.031 m/s (MAD-based 0.027), and the least-squares ego-velocity covariance has a median $\sqrt{\text{tr} \Sigma} = 0.054$ m/s: the radar noise is at the lower end of the 0.05–0.15 m/s range used in the simulations, as the paper states. The rig moves at 1.54 m/s median (2.19 max) and rotates at 0.48 rad/s median, i.e. it is closer to trajectory B than A. The camera

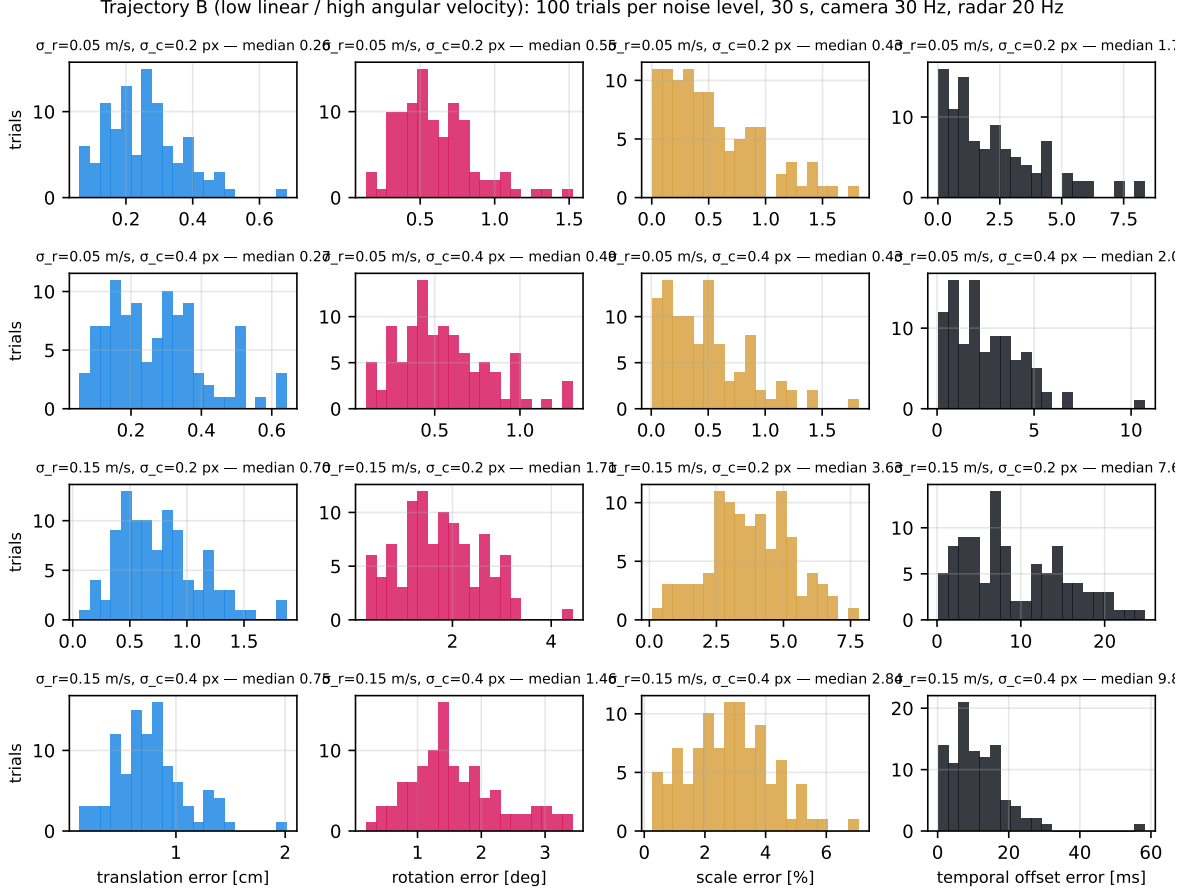


Figure 3: Trajectory B error histograms (paper Fig. 6).

driver left the image header stamps at zero (bag receive times are used), and neither pupil-apriltags nor OpenCV detected the tag36h11 AprilGrid in the calibration bag at 1920×1200 , so the camera intrinsics and an ORB-SLAM3 pose stream, both required for the full spatiotemporal solve, are not available in this re-run.

4 Implementation and Usage

```
cd experiments
for k in 0 1 2 3 4; do
  python sim_studies.py 100 --shard $k --nshards 5 &
done; wait
python sim_studies.py          # merge -> out/sim_studies.json, figures
python real_frontend.py ../../data/handheld/<bag>.bag
python make_tables.py && (cd ../docs && make)
python make_video.py
```

`simulate(kind, ...)` builds a dataset, `Estimator(data).solve(closed_form_init(...))` runs the calibration, `calib_errors` scores it. Hardware path: replace `simulate` by a loader that

Table 2: Closed-form initialisation, reported uncertainty and radar residual per setting (medians).

traj.	σ_r [m/s]	σ_c [px]	closed-form initialisation, median error				reported 1σ , median		radar residual RMS [m/s]
			rot [deg]	trans [cm]	scale [%]	offset [ms]	trans [mm]	offset [ms]	
A	0.05	0.2	0.85	11.1	5.0	45	17.4	1.9	0.038
A	0.05	0.4	1.01	4.5	3.1	50	17.3	1.9	0.039
A	0.15	0.2	2.48	109.4	34.6	45	50.3	3.1	0.127
A	0.15	0.4	2.02	96.9	32.7	50	50.0	2.8	0.130
B	0.05	0.2	0.90	0.4	6.9	80	3.1	1.6	0.038
B	0.05	0.4	0.88	0.4	6.9	95	3.1	1.6	0.038
B	0.15	0.2	2.09	1.8	33.4	65	8.3	2.7	0.127
B	0.15	0.4	2.00	2.0	33.3	90	8.3	2.7	0.128

Table 3: Singular values of the identifiability matrix (8 parameters) for trajectory B over 8 s.

motion	σ_1	σ_2	σ_3	σ_4	σ_5	σ_6	σ_7	σ_8
trajectory B, full excitation	1.07e+01	9.89e+00	6.39e+00	6.02e+00	4.98e+00	2.38e+00	2.23e+00	1.65e+00
rotation about one axis only	8.81e+00	8.72e+00	6.07e+00	4.95e+00	2.31e+00	2.21e+00	1.63e+00	1.60e-16
no motion	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00

provides timestamped scaled camera poses (any monocular SLAM) and radar ego-velocities (`real_frontend.py` shows the latter on the TI driver’s `PointCloud2`).

5 Limitations

- No real-data spatiotemporal calibration: ORB-SLAM3 poses and camera intrinsics are missing (see above); the paper’s Tables I–III are therefore not reproduced.
- Finite-difference Jacobians: 5–10 s per 30-s solve; analytic spline Jacobians would be an order of magnitude faster.
- Trajectory A is weakly excited for translation and offset, and the reported covariance is optimistic there.
- Camera noise is modelled through PnP on a checkerboard as in the paper, not through a SLAM system with drift.
- The prior of the paper’s vehicle experiment (eq. 25) is implemented (`Estimator(prior=...)`) but not exercised.

References

- [1] E. Wise, Q. Cheng, J. Kelly, “Spatiotemporal Calibration of 3D Millimetre-Wavelength Radar-Camera Pairs,” *IEEE Trans. Robotics*, 2023. arXiv:2211.01871.
- [2] E. Wise, J. Peršić, C. Grebe, I. Petrović, J. Kelly, “A continuous-time approach for 3D radar-to-camera extrinsic calibration,” *ICRA*, 2021.
- [3] C. Sommer, V. Usenko, D. Schubert, N. Demmel, D. Cremers, “Efficient derivative computation for cumulative B-splines on Lie groups,” *CVPR*, 2020.
- [4] C. Doer, G. F. Trommer, “An EKF based approach to radar inertial odometry,” *MFI*, 2020.

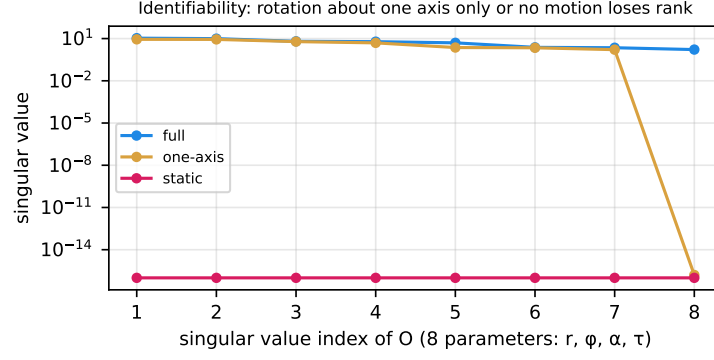


Figure 4: Identifiability: singular values of $\mathbf{O}$ for full excitation, one-axis rotation and no motion.

- [5] C. C. Stahoviak, “An instantaneous 3D ego-velocity measurement algorithm for FMCW Doppler radar data,” M.S. thesis, U. Colorado Boulder, 2019.
- [6] R. Hermann, A. Krener, “Nonlinear controllability and observability,” *IEEE TAC*, 1977.

Handheld rig, radar_camera_imu_velRes_04_v4_0.bag: radar ego-velocity (RANSAC) and IMU rates

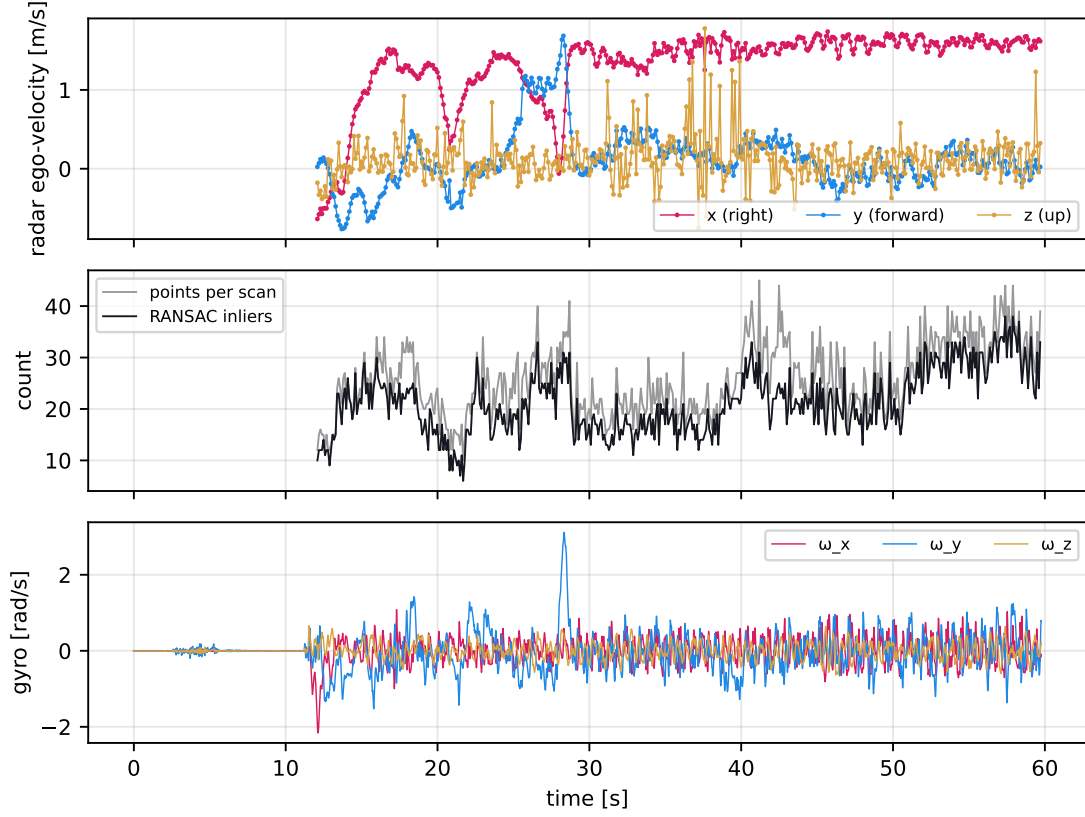


Figure 5: Handheld rig: radar ego-velocity, return/inlier counts and gyro rates.

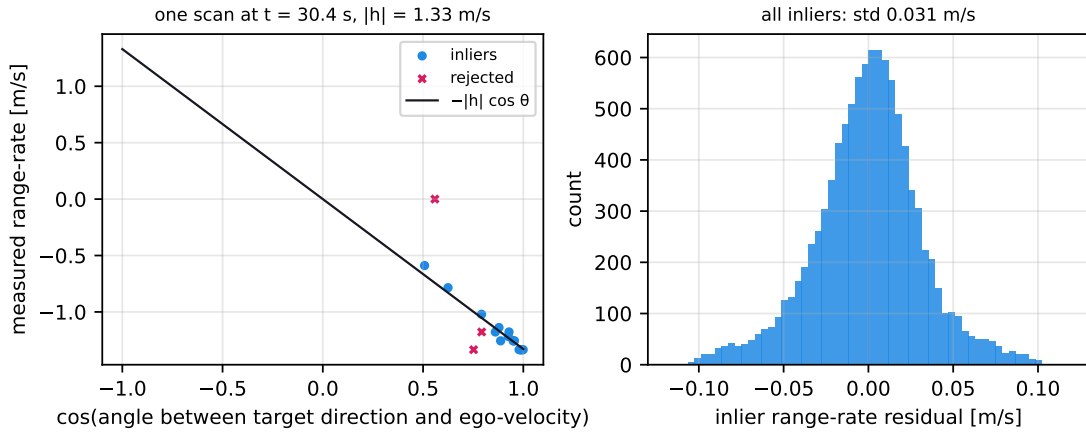


Figure 6: One real scan (range-rate against the cosine of the angle to the ego-velocity) and the inlier residual histogram.